TIO Operations Hub System Design Document

TIO OPERATIONS HUB / CONTROLLED DOCUMENT

TIO-ARC-001 | 2.0 **ISSUED | 24 OCTOBER 2026**

Control field	Controlled value
Document ID	TIO-ARC-001
Version / status	2.0 / Architecture working baseline - review pending
Classification	Internal engineering and operations
Owner / approver	Engineering/architecture owner (proposed) / Security, operations and Product (pending)
Issue date / review	24 October 2026 / On material architecture, security, deployment or data-model change
Source baseline	Repository working tree reviewed 24 October 2026; release notes identify 2.30.0 and in-progress work may be newer
Supersedes	Initial System Design Document dated 24 October 2026

1. Executive summary

TIO Operations Hub is a customer-managed web application for recording and reviewing operational context across clients, projects, environments, products, infrastructure, cases, commitments, contracts, teams and reports. It is implemented as an ASP.NET Core 10 modular monolith with a React/TypeScript client and SQL Server persistence. The supported production shape documented by the repository is one IIS-hosted application instance connected to one SQL Server database, plus configured file storage for attachments and generated exports.

The design prioritizes a single deployable system, relational records, server-side authorization and a controlled DBA-led schema release. It does not depend on service-to-service messaging. The product does not automatically discover or monitor client infrastructure, synchronize Jira, retrieve secrets from vaults, send email, provide customer-facing self-service, or settle financial penalties. Condition, inventory and operational status are user-maintained records. Any proposed integration, hosted multi-tenant model, high availability, public holiday calculation or scale-out must go through a separate architecture decision and security/operability review.

This SDD is based on repository evidence rather than an independent inspection of a production environment. The working tree may contain changes beyond release 2.30.0. Before approving a deployment or external design representation, establish the exact source commit, artifact version, migration set, configuration and evidence baseline.

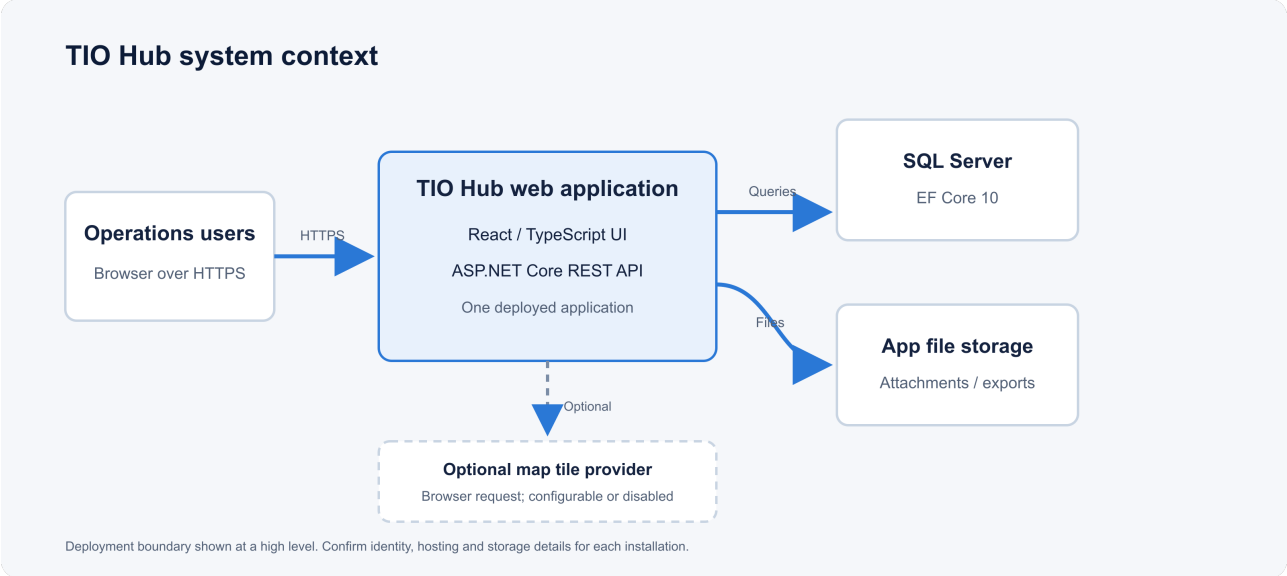
2. Purpose, audience, and scope

This design provides engineering, product, security, deployment and operations reviewers with a coherent system boundary, component model, data/security behavior, runtime topology, key risks and open decisions. It is not a threat model, production as-built, privacy impact assessment, capacity plan, service-level agreement or assurance report.

In scope: application components and modules, request/data paths, SQL and file-backed storage, identity and authorization, reporting workers, release/deployment, recovery assumptions and documented external dependencies.

Out of scope: organization-specific network diagrams, cloud/vendor commitments, production load measurements, detailed per-table schema, exhaustive API contract, final operational procedures, formal threat analysis and legal data-processing roles. The authoritative source for these items is linked in Section 16.

3. System context and trust boundaries



System context: browser UI, IIS/API, SQL Server, file storage and optional map tile provider

The browser loads the SPA and calls the same-origin `/api/v1` service. IIS hosts the ASP.NET Core application and serves built frontend assets. The API authenticates the request, resolves effective user context, applies endpoint/workflow authorization and scope-aware queries, then accesses SQL Server through EF Core. Attachment and export bytes may be stored outside SQL Server in configured directories; database metadata associates files with records and users.

The application does not initiate calls to Jira, client systems, secrets vaults or outbound mail services in the documented design. Jira identifiers are manual references. Optional map tiles are requested directly by the user's browser from the configured HTTPS tile service; public service use, attribution and privacy implications are operator decisions. Bundled geographic data supports the map when tile access is unavailable or disabled.

Trust boundaries: (1) user's browser and local device; (2) IIS/ASP.NET application process; (3) SQL Server; (4) attachment/export/log/backup file stores; (5) organization identity provider/IIS authentication; (6) optional browser-to-tile-provider request. Each boundary requires deployment-specific authentication, network, ACL, logging and retention controls.

4. Functional capabilities and domain scope

Domain	Responsibilities	Important limit
Access and administration	Users, roles, mappings, resource/action grants, settings, audit and supported archive/restore	Administrator status alone does not imply business-record access
Client context	Clients, projects, optional regions, contacts, responsibility boundaries, contextual organization relationships	Organization graph is not an authorization boundary
Technical context	Environments, reported condition, catalog/version/installations, assets/components/storage, network/firewall and dependencies	No automatic discovery or continuous monitoring
Work management	Cases, affected records, impact, state/history, waiting periods, manual Jira references, commitments and escalations	Jira state is not synchronized
Contracts and service measures	Contract terms, renewal, case SLA snapshots, response/resolution evaluation and indicative penalty assessment	Penalty records do not issue money/credits; public holidays are not modeled unless later implemented
Teams and quality	Coverage, handovers, review policies, data-quality evaluation and in-app notifications	Quality results depend on configured rules and record freshness
Reporting	Executive overview, action list, search, report catalog, preview, exports, schedules, history and private presets	CSV/Excel are documented; formatted PDF and email delivery are not

Feature availability is release- and configuration-dependent. The implementation baseline and requirement coverage should be checked before representing any item as generally available.

5. Logical architecture

The backend is a modular monolith: one ASP.NET Core process, one deployable application, logical business modules and one shared EF Core `HubDbContext`. Modules own endpoints, entities and workflow rules but can share context for transactional reads and updates. There are no independent service deployments or message bus in the evidenced architecture.

```
Browser: React 18 / TypeScript SPA
          | same origin, HTTPS, /api/v1
IIS / ASP.NET Core 10 host
  Shared validation, errors, date/time, paging, file/report services
  Identity resolution and authorization context
  Clients | Environment | Catalogue | Inventory | Network/Firewall
  Cases | Commitments | Contracts | Escalations | Teams | Quality
  Reporting | Notifications | Attachments/Notes | Administration
          | EF Core 10 / SQL Server provider
SQL Server: business records, access control, audit, notifications, reporting metadata
          | record/file identifiers and ownership
Configured file storage: attachments, exports, logs and deployment data
```

The React application uses Vite, TanStack Query and React Router and is built into the API's `wwwroot`. The frontend provides navigation and interaction; it is not a security boundary. Every protected operation and file download must enforce access in the API.

6. Runtime components and deployment topology

Runtime component	Function	Operational owner to confirm
IIS / Windows Server	HTTPS binding, hosting integration and configured authentication forwarding	Infrastructure/identity operator
ASP.NET Core 10	API, SPA hosting, validation, authorization, modules, health checks and hosted jobs	Application operations
SQL Server 2019+	Relational system of record and migration history	DBA
File system or approved share	Attachment, export, logs and app data paths	Application/backup operations
User browser	SPA execution and optional map tile requests	Customer endpoint/identity owners
Optional tile endpoint	Browser-loaded map tiles with attribution	Network/map provider owner

Production documentation targets Windows Server 2019 or later, IIS 10, ASP.NET Core Hosting Bundle 10.0.x and SQL Server 2019+ (compatibility level 150+). The application uses its Windows service/app-pool identity for SQL access in the documented deployment pattern. Exact auth mode, identity, TLS, SQL grants, storage paths, tile source and backup policy are installation-specific.

The normal release process builds the frontend and API as one matched package. A DBA reviews/applies the idempotent SQL schema script before matching application files. Production `Database__ApplyMigrationsOnStartup` stays false; readiness checks required migration history and fails closed instead of applying migrations. Development authentication and seeded development records are prohibited in Production.

7. Request, authorization, and data flow

1. The user reaches the HTTPS IIS site and authenticates through the configured Forms or Windows mode.
2. The application resolves the authenticated principal to an active TIO Hub user, groups/mappings where enabled, assigned roles, scope, grants and current session context.
3. Endpoint and workflow checks authorize the requested resource/action. Query filters and domain checks constrain records; archive state is handled separately from scope. Frontend visibility does not replace these checks.
4. The module validates input and applies the business transition. Shared persistence and audit behavior save structured data/history in SQL Server.
5. The SPA renders permitted records. For files, the API checks ownership/authorization before returning bytes from configured storage. Export generation records its job and uses authorized query results.
6. Events and logs capture operational outcomes subject to logging policy. Request logs avoid query strings because search terms can include client data.

The current decision gives business users broad shared client visibility. Resource/action rights govern operations; the policy must not be marketed as per-client or tenant isolation. Administration alone is not business access. Technical detail, export and audit rights are separately controlled. Access changes invalidate cached permission snapshots as documented by the implementation.

8. Data architecture and lifecycle

SQL Server is the system of record for entities, access controls, audit/event history, notifications and reporting metadata. EF Core migrations describe the application model; `database/01-schema.sql` is the production deployment artifact and must be reconciled with migrations. Readiness verifies expected migration entries. No runtime production migration is intended.

Contract-linked cases preserve relevant service-term snapshots so later contract changes do not silently rewrite the case's recorded obligation. Organization relationships belong to a client context and optionally a project; relationships do not confer permission to another client. Deletion and restore workflows require resource/action checks and account for dependent records instead of unrestricted cascading deletion.

Attachments and exports are file-backed. Database metadata, path configuration, application identity ACL, retention and backup must be considered together. Restoring SQL without compatible file data can create records whose files are missing; restoring files without matching database records can leave orphaned content. Logs, backup copies and exports may contain sensitive operational information and require access/retention controls.

9. Authentication and authorization model

The application supports configured Forms or Windows Integrated authentication through IIS. Development sign-in is restricted to Development and Testing. Windows group mappings may contribute effective roles and scopes. Forms credentials are managed as hub accounts; session lifetime and lockout settings are configurable. Verify exact defaults against the release configuration before use.

Effective authorization combines authenticated identity, active hub user, roles, resource/action grants and applicable scope. The API enforces permissions on protected operations. Restore/delete paths and dependent record changes require specific rights according to current decisions and code. Export history/download rules restrict downloads to the generating user; scheduled reports are generated separately for each eligible recipient using that recipient's own access as documented.

Before a production rollout, the owner must approve identity mode, MFA/identity-provider controls, bootstrap administrator, group mapping, lockout/session policy, access review cadence, business visibility boundary, export/technical/audit grants and incident handling. These controls must be validated on the exact target installation.

10. API and presentation layer

The API is REST under `/api/v1`; runtime OpenAPI is at `/openapi/v1.json` for authorized access. Conventions, DTOs, error shapes and current endpoints are maintained in `docs/API.md` and `docs/openapi-v1.json`. The React SPA uses same-origin requests and shared design/navigation components. Static app entry files are handled for freshness while content-hashed assets may be cached.

Security headers, HTTPS redirection and HSTS behavior depend on environment and configuration. Validate IIS forwarding and proxy behavior, cookie settings, CORS/CSP, host allowlists and external links at deployment. The API must independently validate authorization, request fields, time zones, host allowlists, export columns, report filters and workflow transitions.

11. Background processing and reporting

Hosted services evaluate data-quality rules, refresh notification state, sweep SLA obligations and run due report schedules/expiry work. Intervals are configuration-driven; where supported, zero disables recurring work. These jobs run in-process. Multiple application instances may duplicate work or diverge in cache/file behavior unless a later design adds coordination and is verified.

Reporting definitions are centralized and reused for catalog, filters, previews, exports and schedules. Filters are server-validated, queries are access-aware, rows are bounded (documented default maximum 100,000), export

files have configurable storage/retention (documented default 14 days), and downloads are authorized. CSV and Excel are supported in the documented baseline. Schedules create in-hub notifications, not emails. Confirm exact report definitions, calculation denominator, time zone, failure handling and retention before operating a business-critical report.

12. Security, privacy, and threat considerations

Primary assets include client/contact data, support cases, technical inventory/network details, contracts, identity/grants, audit records, attachments, exports, setup credentials, database backups and logs. Relevant threats include compromised user/admin accounts, overbroad grants, cross-scope query defects, unauthorized exports/downloads, malicious uploads, insecure IIS/TLS, SQL overprivilege, path/ACL exposure, stale backups, secret leakage, dependency compromise, tile-provider requests and scheduler duplication.

Baseline controls described in source include server-side authorization, role/action checks, separate access for technical/export/audit behavior, secure cookie/request protections, input/path/host validation, append-only audit guidance, least-privilege SQL permissions, application-controlled file checks, health/readiness checks, release package validation and no production startup migrations. Confirm exact implementation and settings with a release review and test evidence; do not infer a completed threat model or independent assessment.

Before production, complete threat modeling; configuration and penetration testing; IIS/TLS/auth validation; SQL and file ACL review; upload/export abuse cases; secret handling; log/backup protection; privacy/data-flow review; subprocessor/tile review; incident response; vulnerability management; and evidence retention. No SOC 2/ISO certification or external audit report was present in the reviewed material.

13. Reliability, recovery, and operational model

The operations guide defines release, DBA schema application, IIS deployment, health checks, backup, restore and rollback. Startup readiness validates database/schema state before initialization. Health endpoints are `/api/health/live` and `/api/health/ready`; confirm access/monitoring behavior and alert policy locally. Worker failures are logged and generally retried on later intervals, but a retry is not equivalent to durable queueing.

Operators must set service hours, alert thresholds, log retention, backup frequency, attachment/export backup scope, RPO/RTO, recovery tests, capacity and maintenance windows. None are universal product guarantees. Validate a restore in an isolated environment, including SQL and required file content, before switching production.

14. Quality attributes and constraints

Attribute	Design approach	Constraint or acceptance evidence needed
Maintainability	Domain modules within one deployable API and shared context	Keep module ownership explicit; avoid unintended coupling
Access control	API-side roles, scopes and resource/action checks	Negative tests and target configuration review; broad visibility is not tenant isolation
Auditability	Change events, history and export activity	Infrastructure monitoring, retention and independent review require operator controls
Compatibility	Versioned API, EF migrations, matched app/schema package	Upgrade/rollback tests and schema coverage before each release
Portability	Windows/IIS and SQL Server production target	Alternative hosts require separate support/validation
Availability/scale	One application instance is the documented support shape	No availability/capacity SLO; worker/cache/file coordination needed for scale-out
Data freshness	Human-maintained records and reported condition	Review cadence and source ownership required; no live polling
Reporting	Validated server filters, authorized queries, bounded CSV/XLSX	PDF export and email delivery not documented; calculation tests required
Recovery	Database plus file-backed storage restore	RPO/RTO and production-like drill remain owner decisions

No validated production user-count, throughput, latency, database-size or recovery-time measurement is established by this design.

15. Key decisions, open points, and risks

Decisions reflected in current design

- Modular monolith and same-origin deployment.
- SQL Server and EF Core migrations; production schema is DBA-applied.
- Broad shared business visibility with separate action grants; not customer tenant isolation.
- Contextual organization graph does not grant access.
- SLA and penalty calculations are recorded operational assessments; no automatic financial settlement.
- In-process workers and local/configured storage; one application instance remains the supported baseline.

Owner decisions before a production commitment

Decision	Required owner	Why it matters
Identity provider, authentication mode, MFA and group mapping	Security/identity	Account lifecycle, assurance and access behavior
Client confidentiality and access boundary	Product/security/customer owner	Determines whether current shared visibility is acceptable
Hosting, network, capacity and availability target	Infrastructure/service owner	Availability, scale and support scope
Database/file retention, backup, RPO/RTO and restore cadence	DBA/service owner	Recovery and contractual obligations
Tile provider, attribution and browser privacy	Network/privacy owner	External requests and terms
Data controller/processor, jurisdictions and sub-processors	Legal/privacy	Lawful processing and customer notice
Support hours, incident contacts and escalation	Service owner	Operational response responsibility
Assurance target and disclosure policy	Security/legal/executive	Prevents unsupported certification claims

Material risks

- Incorrect auth/IIS/SQL/file configuration can expose data or block valid access.
- Broad visibility may not meet customer confidentiality expectations.
- Database/file restore mismatch can break attachment/export retrieval.
- In-process jobs, caches and file storage do not support unreviewed horizontal scaling.
- User-maintained condition/inventory may be stale or inaccurate.
- Contract terms, time zones, calendar behavior and indicative penalties need business review.
- Working tree and release notes may not align; confirm exact release before deployment.
- Browser map tile service availability/privacy are outside the application boundary.

16. Verification and decision gates

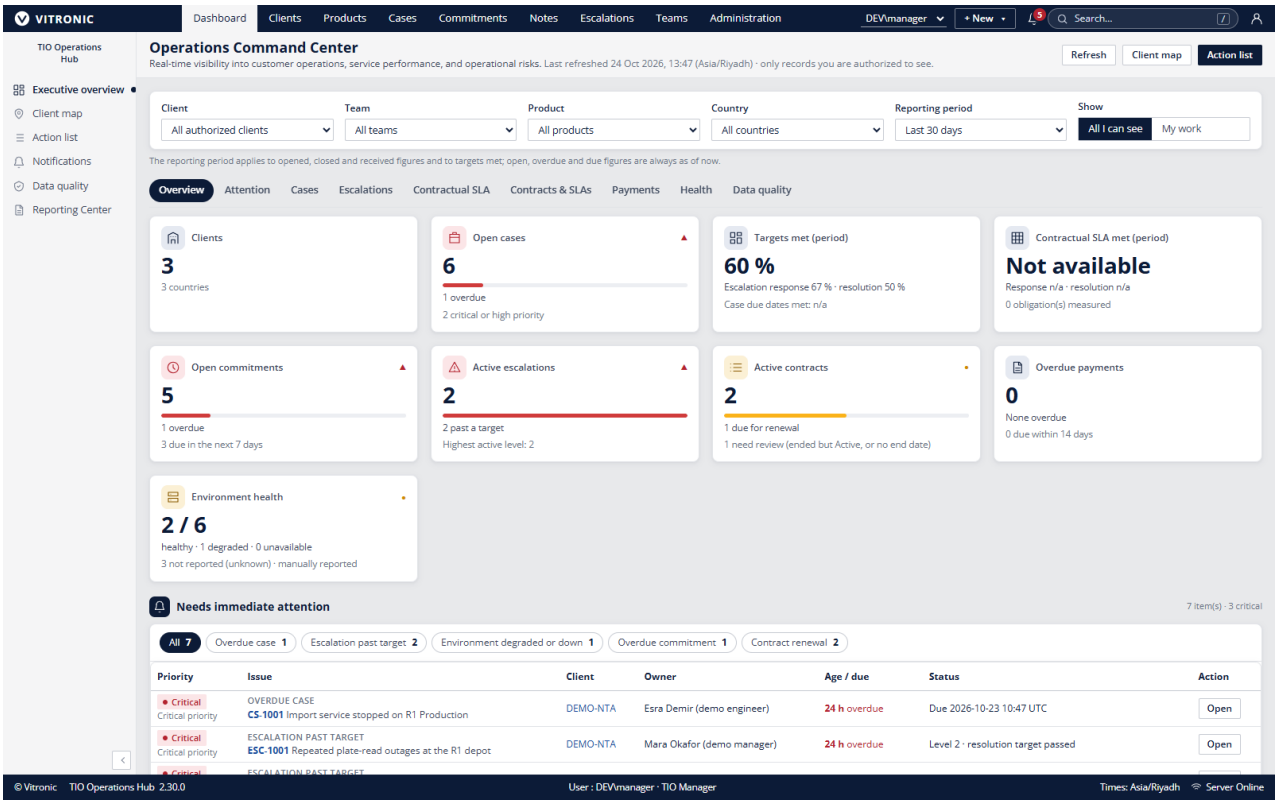
Before release approval: establish source commit and package hash; compare migrations with generated SQL; execute appropriate .NET, frontend and browser tests; review security and permission negative paths; validate build/package contents; install against isolated SQL and target-like configuration; verify health, auth, key workflows and reporting; test backup/restore and rollback; record owner approvals. For production, add customer UAT, environment-specific security review, monitoring, capacity, support, privacy and recovery acceptance. This document does not claim those gates have been completed.

17. Source references

- Project overview and technology
- Decisions and assumptions, including current authorization, organization, SLA, penalty and Reporting Center decisions
- Design rules
- Implementation plan

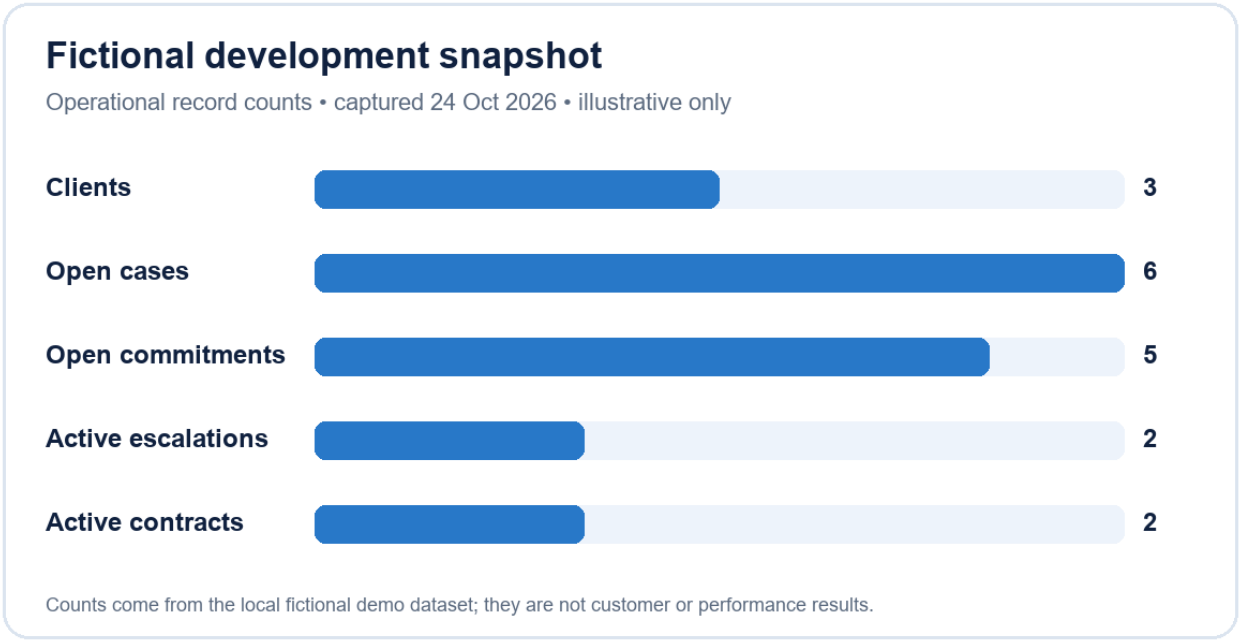
- Requirements coverage
- API guide and OpenAPI document
- Operations and deployment guide
- Release notes 2.30.0 and dated test results
- Organization hierarchy design and stage/decision notes
- Source areas: `src/TioHub.Api/Program.cs`, module registration, security, `HubDbContext`, migrations, `src/web`, `database/01-schema.sql`

Appendix A. Illustrative application interface



Operations overview from local TIO Hub development build 2.30.0

This direct capture uses fictional development data in a dedicated local database. Displayed figures are not customer results or production telemetry. The header includes reference VITRONIC branding; obtain brand-owner approval or replace it with an approved brand-safe image before external distribution.



Counts from the fictional development dashboard

Illustrative record counts only. They are not benchmarks, service-performance measurements or evidence of customer outcomes.

Appendix B. Revision history

Version	Date	Change	Approval
1.0	24 Oct 2026	Initial implementation-oriented SDD	Draft
2.0	24 Oct 2026	Rebuilt with trust boundaries, flows, security, operations, decision gates, and controlled status	Architecture/security review pending